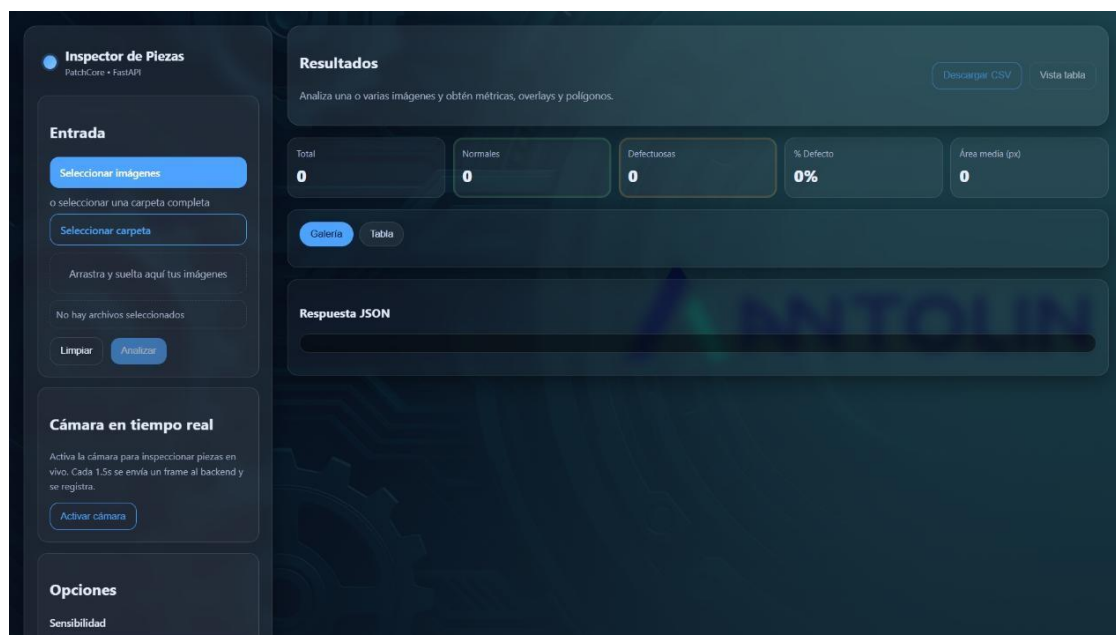


User Manual – Anomaly detection in plastic parts

This document is the user manual of the application, and its purpose is to explain graphically how the application works, as well as to show the wide variety of options that we can modify to customize the application's execution flow. With that said, we will now explain the different options available. This is the first thing we will see when starting the application.



Now, let's go through each option one by one. This is the input section, where we can upload our images to test them in the application. Within this section, we have the **“Select images”** option, which allows us to choose as many images as we want. In addition to that, we also have the option to select a folder so that the application analyzes all the files inside it. Additionally, and for greater convenience, we can also drag and drop the image to upload it.

After uploading the images, we can see the names of all the files we have added, and we have the option to clear the images or to analyze them.

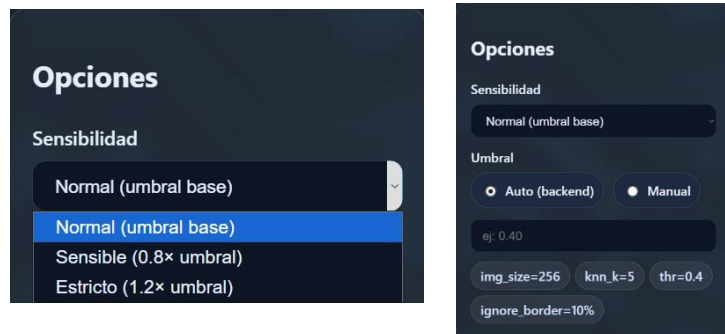


This is the live camera section. As its name suggests, through the “**Enable camera**” option we can inspect parts directly using our device’s camera.

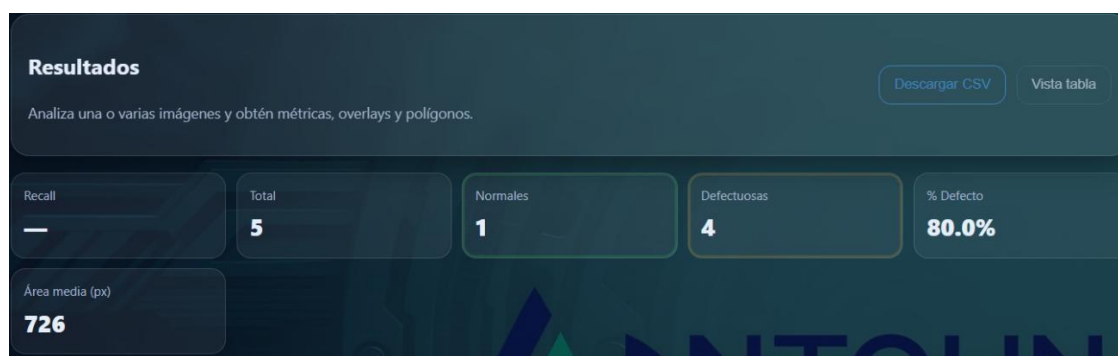


We now move on to the options section, where we can adjust the detection sensitivity threshold as we wish, or alternatively leave it at the default development values.

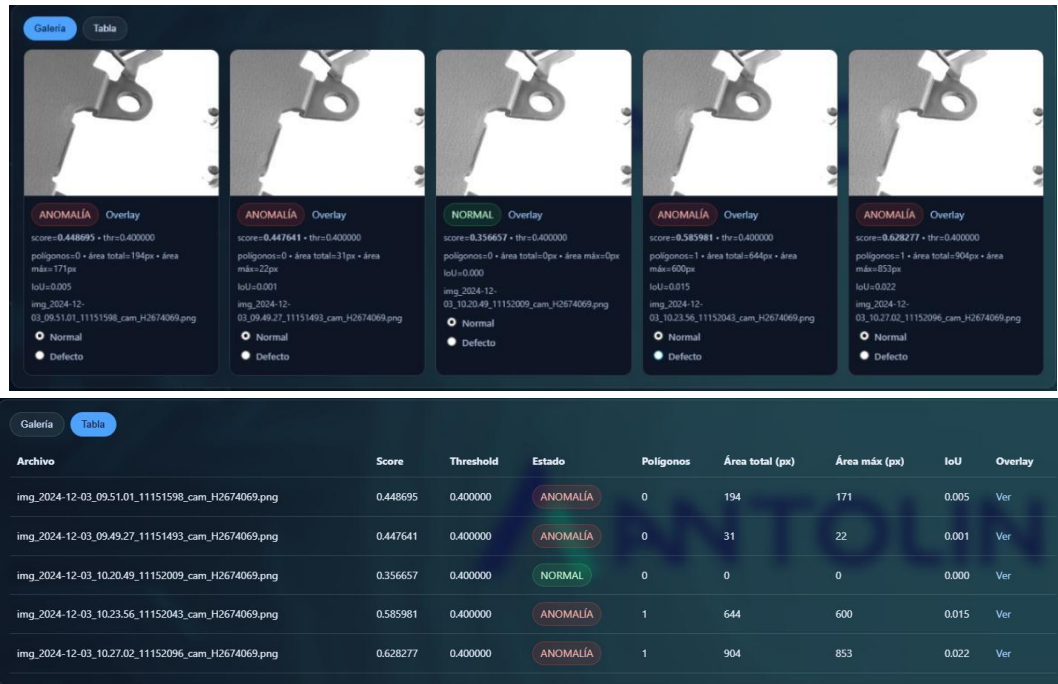
In addition to this, it also includes a dropdown menu with options to increase or decrease the threshold with a single click.



This is the results section. In it, we can see a summary through parameters that reflect our execution. In this case, we have uploaded 5 images to be inspected, of which 1 has been identified as normal and 4 as defective. Additionally, we have an 80% defect rate, and an average area of 726px in the defective regions of the inspected images. As an extra feature, we can download a CSV file that summarizes the execution through tables showing the file name and different metrics related to its identification.



This is the photos section. Here we can view in more detail each of the images we have uploaded to the application, as well as a label indicating their status (anomaly or normal). If we click on the overlay section of any image, we can see an overlay highlighting the inspection area, marking in red any defective zones—if the image contains any. In addition to all of this, we can view different metrics resulting from the analysis of each photo, as well as its name. Finally, we have two views available: “**Gallery**”, which focuses more on the photo preview, and “**Table**”, which focuses on presenting the metrics.



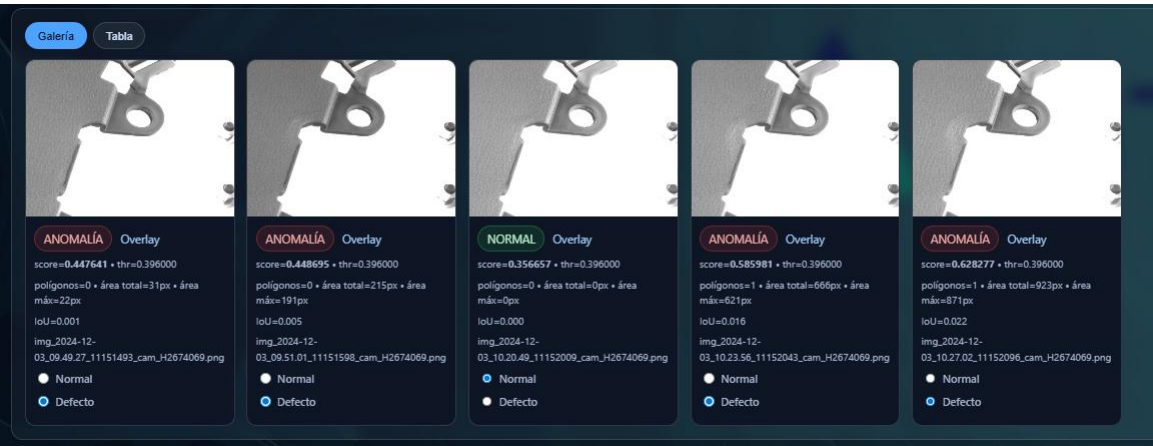
As the final section for more technical aspects, we have the JSON Response. In it, a .json-formatted code is displayed, containing the images we have uploaded along with all their parameters, followed by a matrix of values corresponding to the polygons of each photo—if any exist—along with the parameters used during the application’s execution, such as the threshold used.

```
Respuesta JSON

{
  "threshold": 0.4,
  "results": [
    {
      "idx": 0,
      "filename": "img_2024-12-03_09.51.01_11151598_cam_H2674069.png",
      "score": 0.4486957418483625,
      "threshold": 0.4,
      "is_anomaly": true,
      "total_defect_area_px": 194,
      "max_defect_area_px": 170.5,
      "iou": 0.004661688589804229,
      "overlay_url": "/static/overlays/img_2024-12-03_09.51.01_11151598_cam_H2674069_overlay.png",
      "polygons": []
    },
    {
      "idx": 1,
      "filename": "img_2024-12-03_09.49.27_11151493_cam_H2674069.png",
      "score": 0.44764958094133148,
      "threshold": 0.4,
      "is_anomaly": true,
      "total_defect_area_px": 31,
      "max_defect_area_px": 21.5,
      "iou": 0.0007449058054594386,
      "overlay_url": "/static/overlays/img_2024-12-03_09.49.27_11151493_cam_H2674069_overlay.png",
      "polygons": []
    },
    {
      "idx": 2,
      "filename": "img_2024-12-03_10.20.49_11152009_cam_H2674069.png",
      "score": 0.3566569685935974,
      "threshold": 0.4
    }
  ]
}
```

After running the first analysis on a group of images, the application now allows the user to manually assign the correct ground-truth classification to each inspected image.

This is required to evaluate performance of the anomaly detector and to compute the Recall metric.



Once all the images have been labeled, the user must click the “**Calculate Recall**” button.

The final Recall value, expressed as a percentage, is displayed in the KPI summary section (next to Total, Normals, Anomalies, etc.), updating automatically after calculation.

